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Project Proposal

Emotion-Aware Multi-Artist
Generative Art with AI
Interpretation on Gradient



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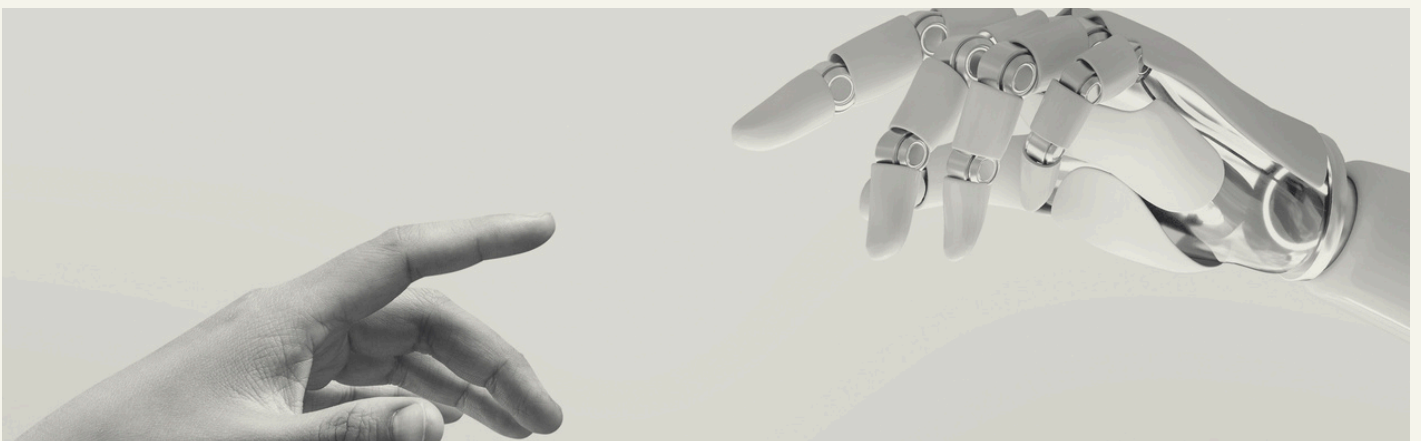
Project Overview

This project develops an AI-powered creative system that generates original artworks inspired by Van Gogh, Monet, and Leonardo da Vinci.

Key Features:

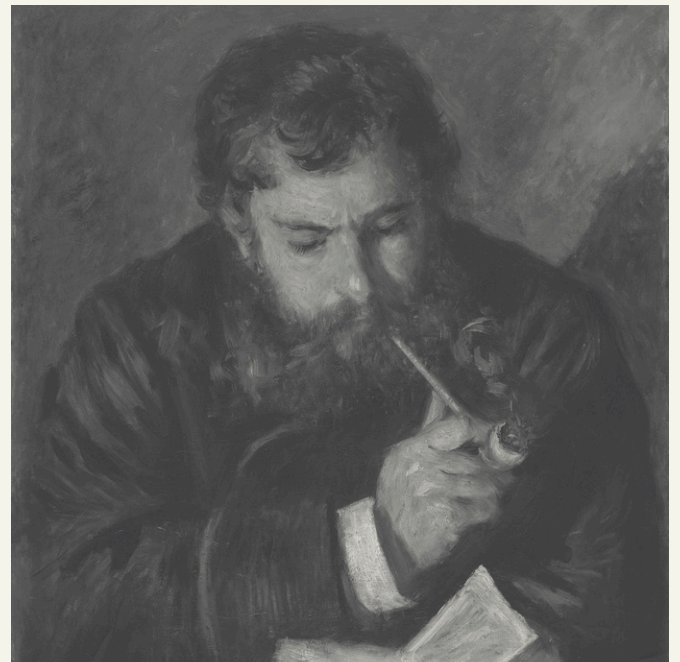
- Emotion Control: Users select Joy, Sadness, or Calm; the AI generates artwork reflecting that emotion.
- AI Interpretation: The system automatically analyzes the generated image and highlights which areas correspond to the selected emotion.
- Multi-Artist Fusion: Combines stylistic features from multiple famous artists in a single output.
- Interactive Demo on Gradient: Users can interact with the model in real-time, adjusting artist style and emotion to instantly see results.

This combination of generative art, emotion modeling, and explainable AI creates a visually impressive and portfolio-ready project for LinkedIn and GitHub.



Objectives

1. Generate high-quality images reflecting the styles of Van Gogh, Monet, and Leonardo da Vinci.
2. Implement user-controlled emotion input for image generation.
3. Develop AI-based interpretation to explain why the generated image reflects the selected emotion.
4. Provide a clean, interactive demo hosted on Gradient.



Dataset

Source

Source: Best Artworks of All Time – Kaggle

- Images resized to 256×256 pixels and normalized.
- Filtered to Van Gogh, Monet, and Leonardo da Vinci
- Emotion labels assigned using CLIP similarity or color-based heuristics (Joy / Sadness / Calm).

Methods

Included

1.Data Preprocessing:

- Filter images by artist
- Resize, normalize, optional augmentation (flip, color jitter)

2.Generative Model:

- StyleGAN2 or DCGAN Multi-Artist
- Latent vectors for artist + emotion vector

3.Emotion Injection:

- Latent vector manipulation according to selected emotion

4.AI Interpretation:

- Map latent vector components to image regions
- Grad-CAM / CLIP similarity → highlight areas contributing to each emotion

5.Interactive Demo on Gradient:

- Dropdown widgets: select artist(s) + emotion
- Generate image + overlay emotion interpretation map
- Output ready for screenshots or direct sharing

Milestones

Milestone	Task
1	Data Collection & Preprocessing (filter artists + emotion labeling)
2	Baseline GAN Training (Multi-Artist)
3	Integrate Emotion Control (3 emotions)
4	AI Interpretation Module (latent → emotion map)
5	Interactive Demo on Gradient + LinkedIn/GitHub Showcase

Expected Outcome

1

High-quality, visually compelling multi-artist artworks

2

Users can control the emotion of the generated image

3

AI explains why

4

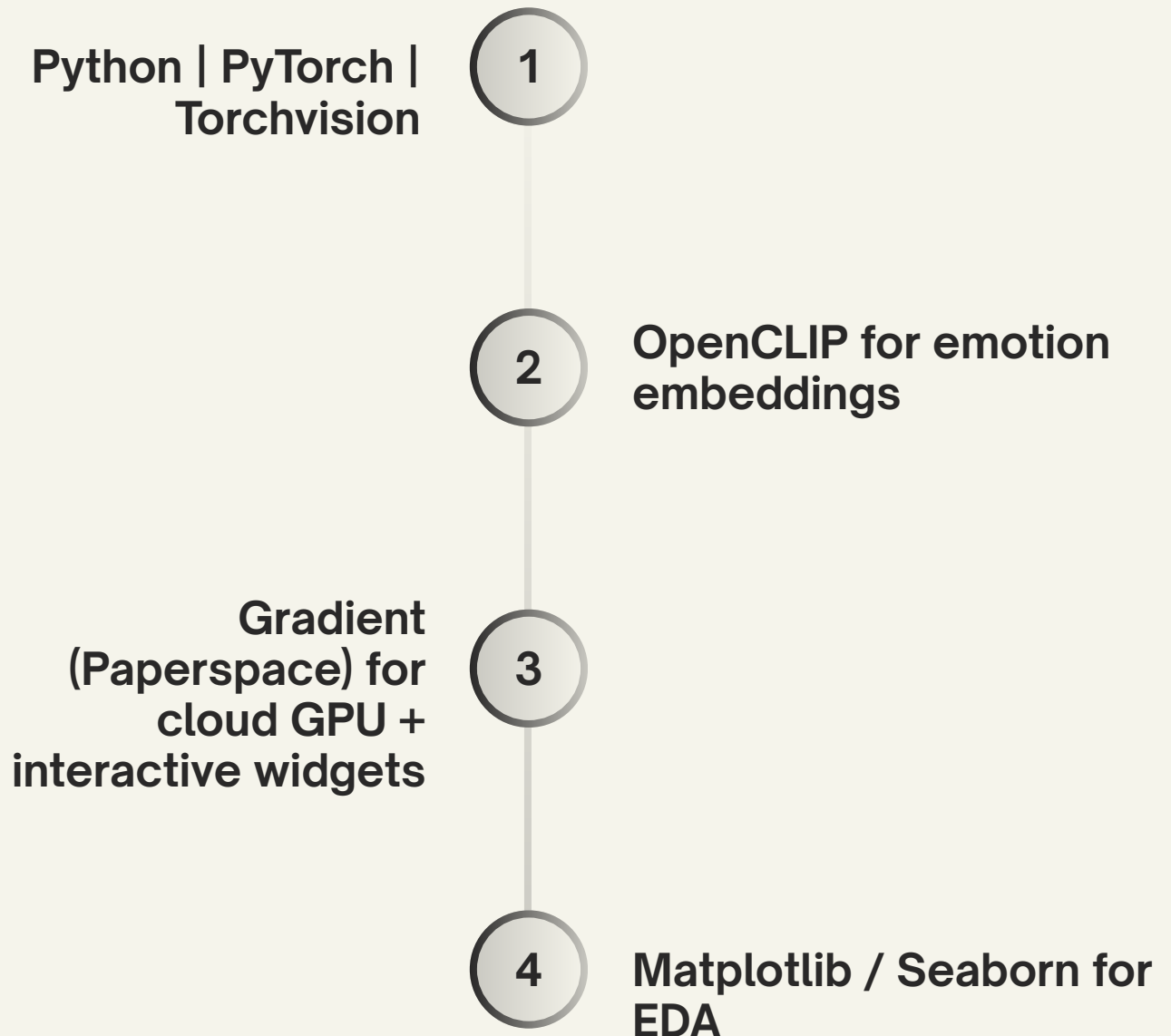
Interactive showcase ready for LinkedIn/GitHub

5

Foundation for future expansion: more artists, continuous emotion scale, higher-res outputs



Tools & Libraries



The End

